

2.  $f(z) = z^3 + az^2 + bz + 175$  where  $a$  and  $b$  are real constants

Given that  $-3 + 4i$  is a root of the equation  $f(z) = 0$

(a) determine the value of  $a$  and the value of  $b$ . (4)

(b) Show all the roots of the equation  $f(z) = 0$  on a single Argand diagram. (2)

(c) Write down the roots of the equation  $f(z + 2) = 0$  (1)



**2.** Given that

$$\begin{aligned} z_1 &= 2 + 3i \\ |z_1 z_2| &= 39\sqrt{2} \\ \arg(z_1 z_2) &= \frac{\pi}{4} \end{aligned}$$

where  $z_1$  and  $z_2$  are complex numbers,

(a) write  $z_1$  in the form  $r(\cos \theta + i \sin \theta)$

Give the exact value of  $r$  and give the value of  $\theta$  in radians to 4 significant figures.

(2)

(b) Find  $z_7$ , giving your answer in the form  $a + ib$  where  $a$  and  $b$  are integers.

(6)

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4. (i) (a) Show that

$$\frac{2 + 3i}{5 + i} = k(1 + i)$$

where  $k$  is a constant to be determined.

*(Solutions relying on calculator technology are not acceptable.)*

(3)

Given that

- $n$  is a positive integer
- $\left(\frac{2+3i}{5+i}\right)^n$  is a real number

(b) use the answer to part (a) to write down the smallest possible value of  $n$ .

(1)

(ii) The complex number  $z = a + bi$  where  $a$  and  $b$  are real constants.

Given that

- $|z^{10}| = 59\,049$
- $\arg(z^{10}) = -\frac{5\pi}{3}$

determine the value of  $a$  and the value of  $b$ .

(4)

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7.

$$f(z) = z^4 + az^3 + bz^2 + cz + d$$

where  $a$ ,  $b$ ,  $c$  and  $d$  are real constants.

The equation  $f(z) = 0$  has complex roots  $z_1$ ,  $z_2$ ,  $z_3$  and  $z_4$

When plotted on an Argand diagram, the points representing  $z_1$ ,  $z_2$ ,  $z_3$  and  $z_4$  form the vertices of a square, with one vertex in each quadrant.

Given that  $z_1 = 2 + 3i$ , determine the values of  $a$ ,  $b$ ,  $c$  and  $d$ .

(6)



2. (a) Express the complex number  $w = 4\sqrt{3} - 4i$  in the form  $r(\cos \theta + i \sin \theta)$  where  $r > 0$  and  $-\pi < \theta \leq \pi$  (4)
- (b) Show, on a single Argand diagram,
- (i) the point representing  $w$
- (ii) the locus of points defined by  $\arg(z + 10i) = \frac{\pi}{3}$  (3)
- (c) Hence determine the minimum distance of  $w$  from the locus  $\arg(z + 10i) = \frac{\pi}{3}$  (3)



7. (i) Shade, on an Argand diagram, the set of points for which

$$|z - 3| \leq |z + 6i|$$

(3)

- (ii) Determine the exact complex number  $w$  which satisfies both

$$\arg(w - 2) = \frac{\pi}{3} \quad \text{and} \quad \arg(w + 1) = \frac{\pi}{6}$$

(6)



5.

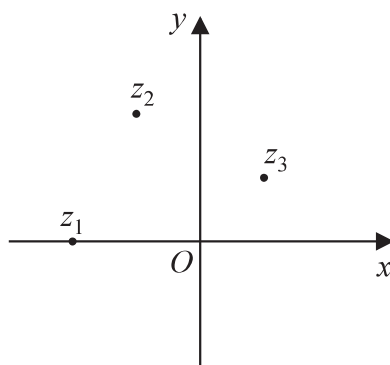


Figure 1

The complex numbers  $z_1 = -2$ ,  $z_2 = -1 + 2i$  and  $z_3 = 1 + i$  are plotted in Figure 1, on an Argand diagram for the complex plane with  $z = x + iy$

- (a) Explain why  $z_1$ ,  $z_2$  and  $z_3$  cannot all be roots of a quartic polynomial equation with real coefficients.

(2)

- (b) Show that  $\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \frac{\pi}{4}$

(3)

- (c) Hence show that  $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(2)

A copy of Figure 1, labelled Diagram 1, is given on page 12.

- (d) Shade, on Diagram 1, the set of points of the complex plane that satisfy the inequality

$$|z + 2| \leq |z - 1 - i|$$

(2)



**3. (a)** Shade on an Argand diagram the set of points

$$\{z \in \mathbb{C} : |z - 1 - i| \leq 3\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leq \arg(z - 2) \leq \frac{3\pi}{4}\right\} \quad (5)$$

The complex number  $w$  satisfies

$$|w - 1 - i| = 3 \text{ and } \arg(w - 2) = \frac{\pi}{4}$$

(b) Find, in simplest form, the exact value of  $|w|^2$

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8. (a) Shade on an Argand diagram the set of points

$$\left\{ z \in \mathbb{C} : |z - 4i| \leq 3 \right\} \cap \left\{ z \in \mathbb{C} : -\frac{\pi}{2} < \arg(z + 3 - 4i) \leq \frac{\pi}{4} \right\}$$

(6)

The complex number  $w$  satisfies

$$|w - 4i| = 3$$

- (b) Find the maximum value of  $\arg w$  in the interval  $(-\pi, \pi]$ .  
Give your answer in radians correct to 2 decimal places.

(2)



10. Given that there are two distinct complex numbers  $z$  that satisfy

$$\{z: |z - 3 - 5i| = 2r\} \cap \left\{z: \arg(z - 2) = \frac{3\pi}{4}\right\}$$

determine the exact range of values for the real constant  $r$ .

(7)

